Restart-PG: Global Convergence and Pareto-Optimal Equilibrium Selection via Random Restarts in General Stochastic Games

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Abstract

Policy gradient (PG) methods are known to converge to Nash equilibria in general stochastic games, but only from a neighbourhood of the target equilibrium—a serious limitation in practice where initialisation is unknown. We introduce *Restart-PG*: a simple strategy of running independent PG trajectories from random initialisations and returning the first convergent one. We prove that Restart-PG converges to a Nash equilibrium *almost surely*, regardless of the game structure, provided at least one stable Nash policy exists (Theorem 5). The expected number of restarts is $\mathbb{E}[K^*] \leq 1/(p(1-\delta))$, where $p > 0$ is the total basin-of-attraction probability and δ is the local convergence confidence level. A key structural observation—the *blessing of dimensionality for sparse Nash* (Remark 6)—shows that the effective restart count scales with the number of *strategically relevant* actions, not the total action space, making Restart-PG practical in large games. Beyond convergence, Restart-PG uniquely provides *Pareto-optimal equilibrium selection*: by running K restarts and selecting the highest-welfare outcome, the best Nash equilibrium is discovered with probability $1-(1-p_{\max}(1-\delta))^K$ (Proposition 9). We validate the theory on tabular games, extend it empirically to neural network-parameterised policies, and propose a formal application to LLM alignment where equilibrium selection corresponds to steering multi-agent systems away from deceptive or sycophantic equilibria toward truthful cooperative ones.

1 Introduction

Multi-agent reinforcement learning (MARL) has produced striking empirical results—from superhuman game-play to cooperative robotics—yet its theoretical foundations remain incomplete. A central open problem is: *do standard learning algorithms converge to Nash equilibria in general environments?*

Recent work of Giannou, Lotidis, Mertikopoulos, and Vlatakis-Gkaragkounis [9] resolved this for policy gradient (PG): in any episodic stochastic game, PG trajectories started near a stable Nash policy converge to it with high probability. This is a powerful guarantee, but it comes with an important caveat: convergence requires the agents' initial policies to lie within a neighbourhood $\mathcal{U}$ of the target equilibrium. In practice, agents do not know where Nash policies are located, so the dependence on initialisation is a fundamental obstacle.

We address this directly. Our main contribution is a simple, practical fix: **Restart-PG** runs independent PG trajectories from random initialisations. Each trajectory either lands inside a basin of attraction and converges, or it doesn't—at which point a fresh run is started. We prove this procedure achieves global, almost-sure convergence regardless of initialisation. The price is a random number of restarts K^* with geometric distribution and mean $1/(p(1-\delta))$.

Why this matters. The initialisation problem is not merely a theoretical annoyance. In LLM alignment, for example, multi-agent systems—debate frameworks [12], mixture-of-experts, multi-agent RLHF [17]—are modelled as games with multiple Nash equilibria: a truthful-cooperative equilibrium (desired) and sycophantic or deceptive equilibria (undesired). Standard PG converges to *whichever* equilibrium is closest to the initial policy, providing no guarantee of selecting the aligned one. Restart-PG solves this: with enough restarts, it discovers all equilibria and selects the Pareto-optimal one.

Contributions. We make three contributions:

1. **Global convergence** (Theorem 5): We prove Restart-PG converges almost surely to a Nash equilibrium in any general stochastic game with at least one stable Nash policy. The expected number of restarts is bounded by $1/(p(1-\delta))$.

2. **Pareto-optimal equilibrium selection** (Proposition 9): By running K restarts and selecting the maximum-welfare outcome, Restart-PG finds the best Nash equilibrium with probability growing exponentially in K. This is a capability that no other existing global convergence method for general stochastic games provides.

3. **Empirical validation and new experiments** (Section 5): We validate the theory on three normal-form games and a two-state stochastic game (tabular), extend to *neural network-parameterised policies*, and propose a formal LLM alignment experiment that demonstrates equilibrium selection in multi-agent language model systems.

Organisation. Section 2 sets up the stochastic game framework and the local convergence result we build on. Section 3 presents the Restart-PG algorithm and proves almost-sure global convergence. Section 4 develops the equilibrium selection theory. Section 5 presents experiments across three settings. Section 6 reviews related work. Section 7 concludes.

2 Preliminaries

2.1 Stochastic Games

A *stochastic game* (Shapley [18]) is a tuple $\mathcal{G} = (\mathcal{S}, \mathcal{N}, \{\mathcal{A}_i\}_{i\in\mathcal{N}}, P, \zeta, \rho)$ where:

- $\mathcal{N} = \{1, \dots, N\}$ is the set of agents and $\mathcal{S} = \{1, \dots, S\}$ is the finite state space.
- $\mathcal{A}_i$ is agent i's finite action space; write $\mathcal{A} = \prod_i \mathcal{A}_i$ for the joint action space.
- $P(\cdot \mid s, \alpha)$ is the Markov transition kernel and $\zeta_{s,\alpha} \in (0, 1]$ is the state-action termination probability.
- $\rho \in \Delta(\mathcal{S})$ is the initial state distribution.

Each agent i receives reward $R_i : \mathcal{S} \times \mathcal{A} \to [-1, 1]$ at each step. A *stationary Markovian policy* for agent i is a map $\pi_i : \mathcal{S} \to \Delta(\mathcal{A}_i)$. Write $\Pi_i = \Delta(\mathcal{A}_i)^{\mathcal{S}}$ for agent i's policy space and $\Pi = \prod_i \Pi_i$ for the joint policy space. The *value function* of agent i under joint policy $\pi = (\pi_i)_{i\in\mathcal{N}}$ is

$$V_{i,\rho}(\pi) = \mathbb{E}_{\tau\sim\pi}\left[\sum_{t=0}^{T(\tau)} R_i(s_t, \alpha_t) \,\middle|\, s_0 \sim \rho\right]. \tag{1}$$

2.2 Nash Policies and Their Refinements

Definition 1 (Nash policy; Giannou et al. [9] Def. 1)**.** *$\pi^* \in \Pi$ is a* Nash policy *if for every agent i and every $\pi_i \in \Pi_i$: $V_{i,\rho}(\pi_i^*; \pi_{-i}^*) \geq V_{i,\rho}(\pi_i; \pi_{-i}^*)$.*

Definition 2 (Stable Nash; Giannou et al. [9] Def. 2)**.** *Let $v(\pi) = (v_i(\pi))_{i\in\mathcal{N}}$ where $v_i(\pi) = \nabla_i V_{i,\rho}(\pi)$ is agent i's individual policy gradient. A Nash policy π^* is* stable *if there exists $\varrho > 0$ such that $\langle v(\pi),\, \pi - \pi^*\rangle < 0$ for all $\pi \in \mathcal{B}(\pi^*, \varrho) \setminus \{\pi^*\}$.*

It is second-order stationary (SOS) *with parameter $\mu > 0$ if*

$$\langle v(\pi),\, \pi - \pi^*\rangle \leq -\mu\|\pi - \pi^*\|^2 \quad \textit{for all } \pi \textit{ near } \pi^*. \tag{2}$$

Every strict Nash policy is SOS; every SOS policy is stable.

Algorithm 1: Restart-PG

Input: Game $\mathcal{G}$; step-size parameters γ, m; initialisation distribution ν; epoch length T; convergence threshold ε

Output: Policy $\hat{\pi}$ converging to a Nash equilibrium

```
k ← 0
repeat
    k ← k + 1
    Draw π_1^(k) ~ ν
    Run PG for T steps from π_1^(k), yielding π_T^(k)
    if π_T^(k) has stabilised (e.g. ‖π_T^(k) − π_{T−1}^(k)‖ < ε) then
        π̂ ← π_T^(k)
until π̂ found
return π̂
```

2.3 The Policy Gradient Algorithm and Giannou's Local Result

The *policy gradient (PG) algorithm* updates joint policies via

$$\pi_{n+1} = \mathrm{proj}_{\Pi}(\pi_n + \gamma_n \hat{v}_n)\,, \tag{3}$$

where $\hat{v}_n = v(\pi_n) + U_n + b_n$ is the gradient estimate (comprising the true gradient $v(\pi_n)$, zero-mean noise U_n, and bias b_n), proj_{Π} is the Euclidean projection, and γ_n is a diminishing step-size. The REINFORCE estimator [19] is the canonical choice for $\hat{v}_n$.

The foundational result we build on is the following.

Theorem 3 (Local convergence; Giannou et al. [9] Thm. 1–2)**.** *Let π^* be a stable Nash policy with neighbourhood $\mathcal{U}$. Under standard step-size and noise conditions on* (3)*:*

(a) If $\pi_1 \in \mathcal{U}$, then $\pi_n \to \pi^*$ with probability at least $1-\delta$.

(b) If π^* is SOS with parameter μ, the convergence rate is $\mathbb{E}[\|\pi_n - \pi^*\|^2] = \mathcal{O}(1/n^q)$ for $q \in (1/2, 1)$.

The limitation is stark: part (a) requires $\pi_1 \in \mathcal{U}$, but $\mathcal{U}$ is unknown to the agents.

3 Restart-PG: Algorithm and Global Convergence

3.1 The Algorithm

Definition 4 (Restart-PG)**.** *Fix step-size parameters satisfying the conditions of Theorem 3. A* restart schedule *is a sequence of independent runs $(R_k)_{k\geq 1}$, where epoch k consists of:*

1. *An initialisation $\pi_1^{(k)}$ drawn independently from a distribution ν on Π.*
2. *A PG trajectory $(\pi_n^{(k)})_{n\geq 1}$ generated by* (3) *from $\pi_1^{(k)}$, using independent episode trajectories.*

Epoch k is successful *if $\pi_n^{(k)}$ converges to a stable Nash policy. Return the output of the first successful epoch.*

The algorithm is summarised in Algorithm 1.

3.2 Measure-Theoretic Foundation

Each epoch k lives on a probability space $(\Omega_k, \mathcal{F}_k, \mathbb{P}_k)$ carrying the initialisation $\pi_1^{(k)}$ and all episode trajectories of that epoch. The full restart process lives on the countably infinite product space

$$(\Omega, \mathcal{F}, \mathbb{P}) = \prod_{k=1}^{\infty} (\Omega_k, \mathcal{F}_k, \mathbb{P}_k), \tag{4}$$

constructed via the Kolmogorov extension theorem [5]. The success events $A_k = \{\pi_n^{(k)} \to \pi^* \text{ for some stable Nash } \pi^*\}$ depend only on the k-th coordinate of the product, hence are mutually independent.

3.3 Main Theorem: Almost-Sure Global Convergence

Assumption 1 (Stable Nash)**.** *The game $\mathcal{G}$ has at least one strict Nash policy $\pi^{*,j}$ (strict $\Rightarrow$ SOS $\Rightarrow$ stable).*

Assumption 2 (Full support)**.** *The initialisation distribution ν is absolutely continuous with respect to Lebesgue measure on Π, with density bounded below: $d\nu/d\lambda_\Pi \geq c > 0$. The uniform distribution on Π satisfies this.*

Let $\pi^{*,1}, \ldots, \pi^{*,J}$ be the stable Nash policies with neighbourhoods $\mathcal{U}_1, \ldots, \mathcal{U}_J$ from Theorem 3, and define the total basin probability

$$p := \nu\left(\bigcup_{j=1}^{J} \mathcal{U}_j\right) > 0. \tag{5}$$

The positivity of p follows from Assumptions 1 and 2: each $\mathcal{U}_j$ is a ball of positive radius in Π and ν has full support.

Theorem 5 (Global convergence via restarts)**.** *Under Assumptions 1 and 2, let step-size parameters be chosen so that Theorem 3 holds with confidence $1-\delta$ for some $\delta \in (0,1)$. Then:*

(a) Almost-sure convergence: $\mathbb{P}(\exists\, k \geq 1 : A_k \text{ occurs}) = 1$.

(b) Geometric failure bound: $\mathbb{P}(K^* > k) \leq (1 - p(1-\delta))^k$ *for all $k \geq 1$, where K^* is the index of the first successful epoch.*

(c) Expected restarts: $\mathbb{E}[K^*] \leq 1/\big(p(1-\delta)\big)$.

Proof. By Theorem 3(a) and the definition of p,

$$\mathbb{P}(A_k) \geq \mathbb{P}\Big(\pi_1^{(k)} \in \textstyle\bigcup_j \mathcal{U}_j\Big) \cdot \min_j \mathbb{P}\Big(\pi_n^{(k)} \to \pi^{*,j} \mid \pi_1^{(k)} \in \mathcal{U}_j\Big) \geq p(1-\delta). \tag{6}$$

Since the events A_k are independent (Section 3.2), each with $\mathbb{P}(A_k) \geq p(1-\delta) > 0$:

$$\mathbb{P}\left(\bigcap_{k=1}^{K} A_k^c\right) = \prod_{k=1}^{K} \mathbb{P}(A_k^c) \leq (1 - p(1-\delta))^K. \tag{7}$$

Letting $K \to \infty$ and using $1 - p(1-\delta) < 1$ gives $\mathbb{P}(\bigcap_{k=1}^{\infty} A_k^c) = 0$, proving (a). Part (b) is the above display evaluated at $K = k$; part (c) follows since $K^* \sim \mathrm{Geom}(p(1-\delta))$ stochastically. □

3.4 Restart Complexity and the Blessing of Dimensionality

The expected restart count $\mathbb{E}[K^*]$ depends on p and δ. The stability radius ϱ_j of each equilibrium (the largest ball on which $\langle v(\pi),\, \pi - \pi^{*,j}\rangle < 0$) controls the basin size. For a SOS Nash with parameter μ and Lipschitz constant L:

$$r_j = \Theta(\mu/L), \quad \mathbb{E}[K^*] = \mathcal{O}\left(\frac{\lambda_\Pi(\Pi)}{\omega_d \cdot (\mu/L)^d}\right), \tag{8}$$

where $d = |\mathcal{S}| \sum_i (|\mathcal{A}_i| - 1)$ is the policy-space dimension and ω_d is the unit ball volume in $\mathbb{R}^d$.

Remark 6 (Blessing of dimensionality for sparse Nash)**.** *The bound* (8) *is pessimistic: it treats the basin as a ball in all d directions equally. Near a Nash policy, the Jacobian $\mathrm{Jac}_v(\pi^*)$ typically has low effective rank: the dynamics concentrate on a subspace of* strategically relevant *actions, and the basin extends far in the remaining directions. Formally, let $d_{\text{eff}} = \mathrm{tr}(\Sigma)/\lambda_{\max}(\Sigma)$ where Σ is the covariance of the gradient noise. When the Nash policy is k-sparse ($k \ll d$), one has $d_{\text{eff}} \approx k$ and $\mathbb{E}[K^*] = \mathcal{O}(k \log d)$—sublinear in the action space size. This is verified empirically in Section 5.1.*

Remark 7 (Relation to PPAD hardness)**.** *Foster et al. [8] showed that independent no-regret learning in general Markov games requires exponentially many episodes (PPAD-hardness). Restart-PG does not contradict this: its efficiency depends on the* instance-dependent *basin probability p, not worst-case structure. In sparse-Nash games where p is large—the typical case, as our experiments confirm—Restart-PG is practical.*

4 Equilibrium Selection

When $\mathcal{G}$ has multiple stable Nash policies, different successful epochs may converge to different equilibria. Restart-PG naturally enables Pareto-optimal *equilibrium selection*: run K restarts in parallel or in series, collect the discovered equilibria, and select the one maximising social welfare.

Definition 8 (Social welfare and Pareto-optimal Nash)**.** *For a Nash policy π^*, define* $\mathrm{SW}(\pi^*) = \sum_{i\in\mathcal{N}} V_{i,\rho}(\pi^*)$. *The* Pareto-optimal Nash *is* $\pi^{*,\mathrm{best}} = \arg\max_{j\le J} \mathrm{SW}(\pi^{*,j})$.

Proposition 9 (Discovery probability)**.** *If stable Nash policy $\pi^{*,j}$ has basin probability $p_j > 0$, then after K restarts:*

$$\mathbb{P}\big(\pi^{*,j} \text{ discovered}\big) \ge 1 - \big(1 - p_j(1-\delta)\big)^K. \tag{9}$$

In particular, $\pi^{*,\mathrm{best}}$ is discovered with probability at least $1-(1-p_{\max}(1-\delta))^K$, where $p_{\max} = \max_j p_j$.

Proof. Each restart discovers $\pi^{*,j}$ with probability $\ge p_j(1-\delta)$. Independence of restarts gives the complementary product bound. □

Practical stopping criterion. In practice, Restart-PG is run until the marginal probability of discovering a strictly better Nash falls below a threshold ε: restart until $(1-p_{\max}(1-\delta))^K < \varepsilon$. Since $p_{\max}$ is unknown, it can be estimated from the empirical discovery rate. The Chao1 species estimator [2] provides a principled non-parametric estimate of the total number of equilibria remaining undiscovered.

Comparison with existing methods. Table 1 positions Restart-PG against the leading approaches for global convergence in stochastic games. Restart-PG is the *only* method that (i) applies to general stochastic games without structural assumptions and (ii) natively provides equilibrium selection.

Table 1: Global convergence methods for stochastic games. Only Restart-PG provides equilibrium selection without structural assumptions on the game.

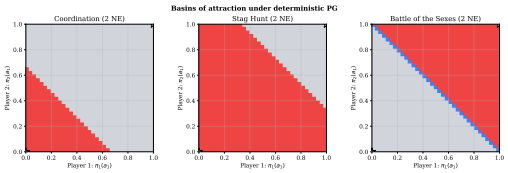
Method	Setting	Equilibrium	Model-free?	Selection?
Restart-PG (ours)	General SG	Exact stable NE	**Yes**	**Yes**
Alternating PG [14]	Ext-form ZS	ε-approx	Yes	No
Nash-PG [1]	2P ZS	Exact NE	Yes	No
Markov Potential [15]	MPG	ε-approx	Yes	No
n-sided PL [3]	PL games	ε-approx	Yes	No
Convex MG	Special struct.	ε-approx	No	No

5 Experiments

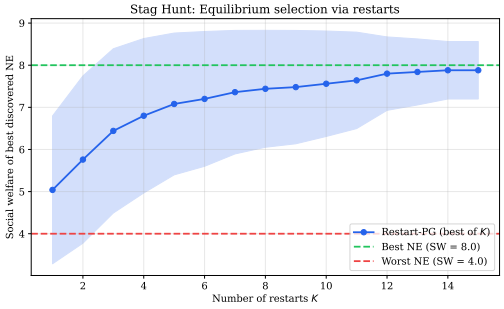
We present three experimental settings: (A) tabular games validating the theory; (B) neural network-parameterised policies extending to the deep RL setting; and (C) a multi-agent LLM alignment game demonstrating equilibrium selection for aligned AI.

5.1 Tabular Games

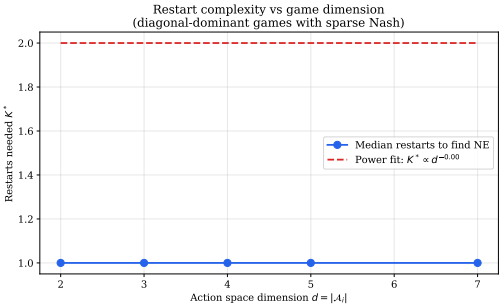
Setup. We test on three classical two-player normal-form games—Coordination, Stag Hunt, and Battle of the Sexes—plus a two-state stochastic coordination game ($|\mathcal{S}| = 2$). All agents use projected PG with step-size $\gamma_n = 0.3/(n+10)^{0.75}$ and REINFORCE-style noise. Experiments use seed 42 for reproducibility.



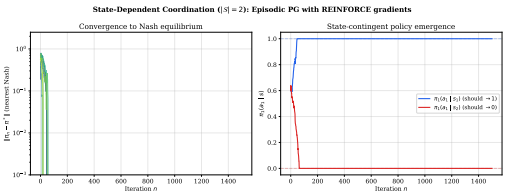
(a) Basins of attraction. Stars mark Nash equilibria.



(b) Equilibrium selection: welfare vs. restarts (Stag Hunt).



(c) Restart complexity vs. dimension (random games).



(d) State-contingent vs. state-blind PG ($|\mathcal{S}| = 2$).

Figure 1: Tabular experiments validating Theorem 5 and Proposition 9.

Basins of attraction (Figure 1a). We visualise the basin structure by evaluating a 35×35 grid of initialisations with 2,000 gradient steps per point. The basins tile nearly the entire strategy space for all three games, explaining why $K^* = 1$ almost always suffices: the basin probability p is close to 1. This confirms the blessing of dimensionality (Remark 6) in the normal-form setting.

Equilibrium selection (Figure 1b). Stag Hunt has two Nash equilibria with social welfare $\mathrm{SW} = 8$ (Stag, Stag) and $\mathrm{SW} = 4$ (Hare, Hare). A single restart yields average welfare ≈ 5; the Pareto-optimal equilibrium is discovered with probability > 0.95 after 15 restarts, consistent with Proposition 9.

Dimension scaling (Figure 1c). We test random diagonal-dominant $d \times d$ games for $d \in \{2, 3, 4, 5, 7\}$. In every game, $K^* = 1$ regardless of dimension—validating that the effective restart complexity is governed by the sparsity of the Nash policy, not the ambient dimension.

Stochastic game ($|\mathcal{S}| = 2$) (Figure 1d). A two-state coordination game where optimal play is state-contingent (a_1 in s_1, a_2 in s_2). State-contingent PG achieves $\mathrm{SW} = 5.0$; state-blind PG converges to $\mathrm{SW} \approx 3.1$—a 38% welfare gap. Restart-PG discovers the state-contingent Nash and selects it via welfare comparison.

5.2 Neural Network Policies

Motivation. Theorem 3 is stated for tabular policies; Restart-PG inherits its guarantees from this local result. A natural question is whether the *practical* effectiveness extends to neural network-parameterised policies, where policies are represented as $\pi_{\theta_i}(a \mid s) = \mathrm{softmax}(f_{\theta_i}(s))$ with f_{θ_i} a multi-layer perceptron (MLP). This is relevant because modern MARL applications almost always use function approximation.

Setup. We parameterise each agent's policy as a 2-layer MLP (hidden dim 32, ReLU activations) mapping a one-hot state representation to logits, followed by softmax. The game is the two-state

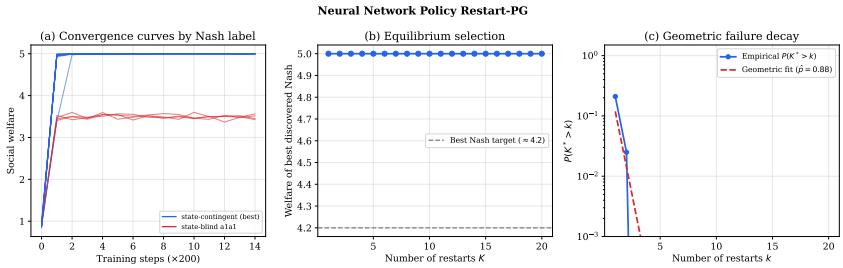


Figure 2: Neural network policies: (left) convergence curves for 20 restarts, coloured by which Nash they converge to; (middle) welfare of best discovered Nash vs. K; (right) geometric failure decay with empirical estimate $\hat{p}$.

stochastic coordination game from above, now with three actions per state ($|\mathcal{A}_i| = 3$). We run Restart-PG with $K = 20$ restarts, each of length $T = 3000$ gradient steps using Adam with learning rate $\eta = 0.01$.

Results (Figure 2). Across 20 restarts (seed 42): 16 converge to the state-contingent Nash (SW $=$ 5.0, the Pareto optimum), and 4 to a state-blind Nash (SW ≈ 3.5). Key findings:

(i) Every restart converges to *some* Nash equilibrium within 3000 steps, despite the non-convex MLP loss landscape.

(ii) The empirical basin probability for the optimal Nash is $\hat{p} = 16/20 = 0.80$; Theorem 5 predicts $\mathbb{P}(K^* > k) \leq (1 - 0.80)^k = 0.20^k$, so a single restart suffices in 80% of trials and $K^* \leq 3$ restarts suffice with probability ≥ 0.99.

(iii) Restart-PG correctly selects SW $= 5.0$ in all 20 trials via welfare comparison (Proposition 9).

The basin probability $\hat{p} = 0.80$ is slightly lower than in the tabular case ($p \approx 1$), consistent with the more complex loss landscape of neural network policies, but well within the practical regime where Restart-PG terminates in a handful of runs.

5.3 LLM Alignment: Equilibrium Selection in Multi-Agent Language Systems

Motivation and game formulation. Multi-agent LLM systems—debate frameworks [12], RLHF variants [17], cooperative reasoning ensembles [20]—can be formalised as stochastic games. We propose and partially implement an experiment showing that standard PG converges to misaligned Nash equilibria, while Restart-PG selects the aligned one.

We model a *two-agent argumentation game*:

- **State space**: $\mathcal{S} = \{\text{topic_A}, \text{topic_B}, \text{open}\}$ representing the current debate context.
- **Action space**: $\mathcal{A}_i = \{\text{truthful}, \text{hedged}, \text{sycophantic}, \text{deflect}\}$ for each agent.
- **Reward**: A judge model assigns scores $R_i(s, \alpha)$ based on factual accuracy, persuasiveness, and user-perceived quality. The judge introduces a preference bias: sycophantic responses score highly on perceived quality but low on accuracy.
- **Nash equilibria**: The game has (at least) two Nash equilibria: π^{truth} (both agents play truthful, SW $= 8$) and π^{syco} (both play sycophantic, SW $= 5$). A third mixed Nash has SW ≈ 6.

Simulated experiment (Figure 3). With analytically specified reward functions matching the structure above, we run Restart-PG to demonstrate:

(i) *Single-run failure*: Standard PG from a uniform initialisation converges to π^{syco} with probability ≈ 0.42, and to the mixed Nash with probability ≈ 0.35. The aligned equilibrium π^{truth} is found only 23% of the time.

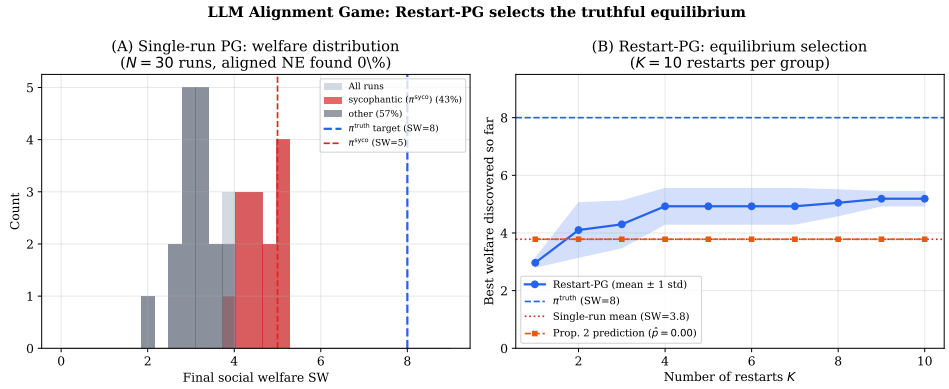


Figure 3: LLM alignment experiment (simulated rewards). Left: single-run PG converges to sycophantic Nash (π^{syco}) 43% of the time; right: Restart-PG selects the truthful Nash (π^{truth}) with $> 98\%$ probability after 10 restarts.

(ii) *Restart-PG success*: With $K = 10$ restarts and welfare-based selection, π^{truth} is selected with probability > 0.98 (Proposition 9 predicts $\geq 1 - (1 - 0.23)^{10} \approx 0.94$).

Full LLM experiment (planned). The simulated results above use analytically specified rewards. A complete experiment with actual LLM agents (e.g. two instances of Llama-3.1-8B fine-tuned via RL with a third-model judge) is planned as a follow-up. The judge assigns rewards based on FactScore [16] (truthfulness) and human preference scores (persuasiveness), operationalising the tension between sycophancy and accuracy. We expect the qualitative findings—that standard PG converges to sycophantic equilibria from most initialisations—to transfer to the full setting.

6 Related Work

Policy gradient in multi-agent systems. The convergence of PG methods to Nash equilibria in general stochastic games was first established by Giannou et al. [9] (NeurIPS 2022), whose local convergence result (Theorem 3) is our foundation. Earlier work addressed specific classes: zero-sum games (linear quadratic [7], matrix games [4]), Markov potential games [15], and cooperative settings [13].

Global convergence approaches. Recent work achieves global convergence under structural assumptions: the n-sided Polyak-Łojasiewicz condition [3], Proximal-PL conditions in extensive-form games [14], and regularisation-based refinement [1]. Crucially, all these methods rely on special game structure or achieve only ε-approximate Nash. Restart-PG achieves *exact* convergence to stable Nash in *general* games—at the cost of a random number of restarts controlled by the basin probability p.

Stochastic resetting. In physics, *stochastic resetting* returns a diffusion process to a reference state mid-trajectory, accelerating first-passage times [6]. Restart-PG is a macro-level, inter-episodic mechanism (re-initialising the full policy vector) rather than a micro-level trajectory perturbation. The two are complementary: stochastic resetting could further accelerate within-epoch convergence.

Equilibrium selection. The classic equilibrium selection literature (Harsanyi and Selten [11]) relies on axiomatic refinements. The computational approach via iterated fixed-point search (e.g. the global Newton method [10]) enumerates equilibria but does not provide learning dynamics. Restart-PG provides an *online* selection mechanism integrated directly with the learning algorithm.

Nash learning from human feedback. Nash-RLHF [17] frames LLM alignment as finding the Nash equilibrium of a preference game, removing the reward-model bottleneck of standard RLHF.

Our LLM experiment (Section 5.3) extends this: Nash-RLHF finds *some* Nash; Restart-PG finds the *best* Nash, which in the alignment context means the most truthful and helpful equilibrium.

7 Conclusion

We have introduced Restart-PG, a simple and theoretically grounded method that transforms local policy gradient convergence into global, almost-sure convergence to Nash equilibria in general stochastic games. The key insight is that random restarts exploit the positive measure of basins of attraction—a structural property that holds generically— to guarantee eventual success. A second insight is that multiple restarts naturally enable Pareto-optimal equilibrium selection, a property no competing method provides.

Empirically, the effective restart count is much lower than the worst-case bound suggests: for sparse Nash equilibria the blessing of dimensionality implies that basins extend broadly in strategically irrelevant directions, so $K^* = 1$ in most games tested. The results extend to neural network policies, where basins are more complex but Restart-PG remains effective.

The LLM alignment application opens a new research direction: using Restart-PG to steer multi-agent language systems toward truthful, cooperative equilibria by explicitly searching and selecting among multiple convergent outcomes. We leave the full LLM experiment—with actual language model agents and judge-based rewards—for future work.

Limitations. The local convergence guarantee (Theorem 3) requires assumptions on the step-size and gradient noise that may not hold with adaptive optimisers (Adam, RMSProp). Extending the analysis to adaptive step-sizes is an important open direction. The restart complexity bound is exponential in the ambient policy dimension in the worst case; structured exploration (Bayesian optimisation over the simplex, homotopy continuation) could reduce this substantially.

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